MULTI-AGENT AI SYSTEMS

Enterprise Multi-Agent Workforce: Architectural Blueprint

1. Multi-Agent Systems Paradigm

Multi-agent systems decompose complex enterprise workflows into specialized, collaborating autonomous agents. Each agent operates with tailored system prompts, dedicated toolsets, and scoped responsibility.

2. Workforce Agent Hierarchy

Agent Role	Primary Responsibility	Input Artifact	Output Artifact
Supervisor / Orchestrator	Task decomposition, delegation, and dependency tracking	User Objective	Execution Plan & Final Synthesis
Research Specialist	Web search, API queries, and document extraction	Query Sub-task	Raw Fact Dossier
Quantitative Analyst	Statistical computation, data transformation, and modeling	Raw Data	Structured Tables & Insights
Technical Writer	Drafting clear, executive-ready documentation	Analyzed Data	Formatted Reports
Quality Auditor	Fact-checking, hallucination detection, and compliance check	Draft Report	Validated Final Deliverable

3. Inter-Agent Communication Protocol

- Uses LangGraph / CrewAI shared state dictionary.
- Memory passes strictly validated JSON schemas to prevent context drift and hallucinations.